

APPENDIX K — COMMUNICATIONS, SYNCHRONIZATION, AND DATA PIPELINE IMPLEMENTATIONS (NON-LIMITING)

A. Overview

This appendix provides non-limiting examples of communications, synchronization, and data pipeline implementations for transferring data from one or more nodes to a computing system and producing privacy-preserving outputs. Implementations may be on-node, on a mobile device, on a remote server, or distributed.

B. Communications Links (Non-Limiting)

In various embodiments, node-to-computing communication may include one or more of: 1. Bluetooth Low Energy (BLE) for periodic upload, streaming, and pairing workflows. 2. Near-field pairing or proximity-assisted pairing for initial device association. 3. Wired transfer via docking base or cable in certain embodiments. 4. Indirect transfer via local storage followed by later synchronization. In certain embodiments, communications are authenticated and encrypted.

C. Time Synchronization (Non-Limiting)

In various embodiments, time alignment between nodes is achieved using one or more of: 1. time-stamp synchronization at pairing and at periodic intervals, 2. drift estimation and correction, 3. event-marker alignment using synchronized triggers, and/or 4. correlation-based alignment using shared context signatures (e.g., motion events observed by multiple nodes). In certain embodiments, the system stores a time-offset model per node and updates it as new data is collected.

D. Data Pipeline Stages (Non-Limiting)

In various embodiments, the data pipeline includes one or more stages: 1. acquisition and buffering on-node, 2. preprocessing (filtering, normalization, drift correction), 3. quality estimation (quality indices, contact integrity checks, motion gating), 4. feature extraction (time-domain, frequency-domain, and state features), 5. metric generation (privacy-preserving indices, zones, curves), 6. correlation computation (cross-device and/or cross-user), and 7. reporting via privacy-preserving interface outputs and optional export.

E. Storage and Retention Modes (Non-Limiting)

In various embodiments, storage may include one or more of: raw-signal caching for short durations for processing only, derived-metric storage for long-term trends, local-only storage mode, and/or consent-gated storage where certain derived metrics are eligible for exchange. In certain embodiments, retention policies are user-configurable and support deletion of cached raw segments.

F. On-Device vs. Cloud Processing Variants (Non-Limiting)

In various embodiments: 1. on-device processing computes metrics locally and uploads only derived representations, 2. cloud-assisted processing uploads validated windows (raw or compressed) subject to privacy settings, and 3. hybrid processing computes quality indices on-node, metrics on-device, and correlation in a server environment under consent gating.

G. Compression and Efficient Transfer (Non-Limiting)

In various embodiments, transfer efficiency is improved by: window-based uploads rather than continuous streams, compression of raw or feature representations, event-triggered upload policies, and adaptive sampling and duty cycling with quality gating.

H. Reliability and Fault Handling (Non-Limiting)

In various embodiments, the system includes: 1. buffering and retransmission mechanisms, 2. partial upload recovery, 3. integrity checks using checksums or message authentication, and 4. fallback behaviors when connectivity is unavailable, including delayed synchronization.

I. Variations

Any communications, synchronization, storage, and processing architecture described herein may be substituted with an equivalent architecture while maintaining the functional intent of longitudinal capture, validity gating, privacy-preserving metric generation, and correlation under consent.